

Entanglement classification with quantum reservoir computing

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Abstract

Reservoir Computing constitutes a cost-efficient alternative to Recurrent Neural Networks that has been shown to work well for tasks such as time series prediction. In recent times, the use of quantum systems as reservoirs has allowed to efficiently solve purely quantum tasks with this machine learning scheme. One of these tasks is entanglement classification. In this work, we use a spin network as a reservoir, and study how it performs at classifying the entanglement of two-qubit states. In particular, we pay special attention to the impact of the internal memory of the reservoir in the accuracy of the classification for different families of states.

1 Introduction

The basic idea of Reservoir Computing (RC) is to replace the deep hidden layers in a Recurrent Neural Network (RNN) with a complex physical system that automatically processes the input information through its evolution, without the need to optimize any weights. This system is called reservoir. Although this approach doesn't work for every machine learning task, it has been shown to function really well for time series prediction. The advantage lies in the training process, which is reduced to solving a linear system of equations to find the optimal weights for the output layer, as opposed to the computationally demanding backpropagation methods used in RNNs.

The RC scheme consists of three basic layers. First, an input layer, where the information is encoded into some degrees of freedom of the reservoir. Then, the reservoir layer, where the information is processed via the natural evolution of a physical system. If the system consists of multiple nodes, its evolution in time will naturally include recursion, which is the key element that allows RNN to retain information about multiple past inputs. That is, the reservoir will have memory, which explains its usefulness in time series prediction. Additionally, the reservoir should have the following three properties [4]:

- Non-linear processing: the reservoir should transform the input non-linearly.
- Separability: different inputs should be processed to different outputs.
- Fading memory: the performance of the algorithm should be independent of the initial conditions of the reservoir, that is, its state before inserting the input. Thus, the reservoir should only have short-term memory, forgetting about its long-past states.

Finally, we have an output layer, where some quantities x_i of the reservoir are measured. The final output, Y , of the scheme is usually obtained as a linear combination of these quantities. The weights of this linear combination are contained in a matrix of coefficients, W . During the training step, we modify W using linear regression so that the outputs of the scheme match the desired values. This training procedure is very simple and efficient compared to the usual RNN backpropagation strategy.

RC can be performed using a wide range of physical systems as reservoirs. The ultimate goal is to choose a system that will map the input into a much bigger state space, allowing the machine to clearly separate the information, distinguish between different inputs, and learn effectively as a consequence. Quantum systems satisfy this requirement particularly well, as the dimension of the Hilbert space of a system of N spin particles grows exponentially with the number of them. For spin-1/2 particles, this is 2^N . The fact that the degrees of freedom of a quantum reservoir increase exponentially with its size means that this type of reservoir will require less nodes in order to outperform other reservoir systems. Besides that, quantum reservoirs allow to solve both classical

and purely quantum tasks, as they can take quantum states as inputs. The goal of this project is to study the application of Quantum Reservoir Computing (QRC) to classifying the entanglement of a two-qubit quantum state. The code developed can be found in the following repository: https://github.com/estaffet/QRC_entanglement_classification.git

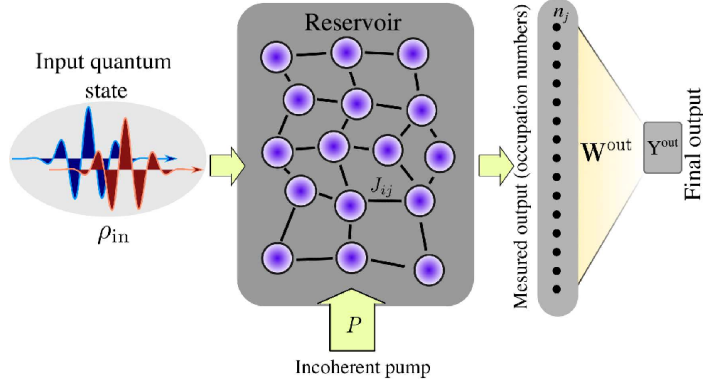


Figure 1: Quantum Reservoir Computing scheme. Taken from [2].

2 Theoretical model

2.1 Ising model

It has been shown that fully-connected transverse-field Ising lattices work well as quantum reservoirs. These are described by the following Hamiltonian [3]:

$$H = \sum_{i,j} J_{ij} X_i X_j + h \sum_i Z_i, \quad (2.1)$$

where X_i and Z_i represent the spin operators along the x and z directions, respectively, for the i -th spin of the lattice. For spin-1/2 particles, these operators are the Pauli matrices. The coupling matrix $(J)_{ij}$ is random, and allows for interactions between any pair of spins, as opposed to the usual nearest neighbour approximation. Its entries are randomly drawn from the uniform distribution in the interval $[0,1]$.

Let's first discuss the common use case of time series prediction. If we wanted to analyze a sequence $u = (u_0, u_1, \dots, u_k, \dots)$, at each time step $t = k\Delta t$ (with $k = 1, 2, 3, \dots$) the state of the first node in the lattice would be replaced by $|\psi_k\rangle = \sqrt{1 - u_k} |0\rangle + \sqrt{u_k} |1\rangle$. This is achieved by taking the partial trace over all spins except the first one. Between each time step, the reservoir is left to evolve naturally. In this way, the state of the reservoir at the k -th step will be influenced by all the previous elements in the sequence. Thus, the reservoir processes the entire sequence simultaneously, keeping memory of its past states, which explains its usefulness in time series predictions [1].

In our present case, the target task is not temporal, meaning that the memory of the dynamical system is not required in principle. The input data will be two-qubit quantum states, and the targets will either be binary classification labels (entangled/separable) or some numerical entanglement measure that characterizes the states. This type of scheme is known as Extreme Learning Machine (ELM). However, inspired by the time series prediction protocol, it is plausible that reinserting the same input multiple times at different time steps might improve the performance of this classification algorithm, especially for complex quantum states. For the purpose of evaluating the impact of input reinsertion, we will start working with Werner states. These are two-qubit states that are parametrized by a single parameter. For this reason, its classification in the training process should be simple enough for the reservoir to not need reinsertion. Then, we will study more complex families of two qubit-states, dependent on more than one parameter. The goal is to analyze if reinsertion improves classification in these cases, and determine the optimal number of reinsertions.

2.2 Approach

First, the reservoir is initialized to a random state, ρ_0 . Then, we take the partial trace over all spins except the first two, and inject the input state, ρ_{input} , there. Explicitly, the reservoir is left in the following state:

$$\rho = \rho_{input} \otimes \text{Tr}_{1,2}\{\rho_0\}. \quad (2.2)$$

where the notation Tr_x indicates the partial trace over all spins except those in the subindex. Next, the reservoir is left to evolve for a time Δt following the dynamics of the Hamiltonian, obtaining:

$$\rho(\Delta t) = e^{-iH\Delta t}[\rho_{input} \otimes \text{Tr}_{1,2}\{\rho_0\}]e^{iH\Delta t}. \quad (2.3)$$

After this time interval, we extract some data from the reservoir by measuring different observables O_j over each spin:

$$x_j = \text{Tr}[O_j \rho(\Delta t)]. \quad (2.4)$$

This data is then linearly combined with weights w_i to produce the final scalar output. These weights are optimized in the training step to best match the desired outputs for the set of training inputs. In our case, we measured X_i , Y_i and Z_i over each of the spins. In practice, in the training step we repeat the mentioned process with multiple inputs, building a matrix of observables:

$$X = \begin{pmatrix} \text{Tr}[O_1 \rho^{(1)}(\Delta t)] & \text{Tr}[O_2 \rho^{(1)}(\Delta t)] & \dots & \text{Tr}[O_k \rho^{(1)}(\Delta t)] & 1 \\ \text{Tr}[O_1 \rho^{(2)}(\Delta t)] & \text{Tr}[O_2 \rho^{(2)}(\Delta t)] & \dots & \text{Tr}[O_k \rho^{(2)}(\Delta t)] & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \text{Tr}[O_1 \rho^{(M)}(\Delta t)] & \text{Tr}[O_2 \rho^{(M)}(\Delta t)] & \dots & \text{Tr}[O_k \rho^{(M)}(\Delta t)] & 1 \end{pmatrix}, \quad (2.5)$$

where the superscript that ranges from 1 to M labels the different input states used for training. Additionally, an extra column has been included in the matrix to account for the independent term of the linear regression:

$$Y = XW \quad (2.6)$$

where Y are the targets of the training step, and W is a vector of weights.

If we wanted to reinsert the input, we would repeat the previous steps iteratively by taking the partial trace at the end of each evolution interval, and end up with the following state:

$$\rho(k\Delta t) = e^{-iH\Delta t}[\rho_{input} \otimes \text{Tr}_{1,2}\{\rho((k-1)\Delta t)\}]e^{iH\Delta t}. \quad (2.7)$$

3 Results and discussion

3.1 Werner states

The first goal was to classify the entanglement of uniparametric 2-qubit states known as Werner states. These are given by the following expression:

$$\rho_W(\lambda) = \lambda |\psi^-\rangle \langle \psi^-| + \frac{1-\lambda}{4} \mathbb{I}, \quad (3.1)$$

where

$$|\psi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \quad (3.2)$$

is a Bell state (maximally entangled), and $\lambda \in [-1/3, 1]$. These states are separable for $\lambda \leq 1/3$ and entangled for $\lambda > 1/3$. Since this family of states depends on a single parameter, one insertion of the input should be enough for the reservoir to classify them correctly. In fact, since this task should be fairly easy for a quantum reservoir, the goal was to predict directly the value of the parameter λ given an input Werner state. The following figure shows the predictions in the training and testing steps. Since the estimation is almost perfect, the target and predicted values overlap in the graphs.

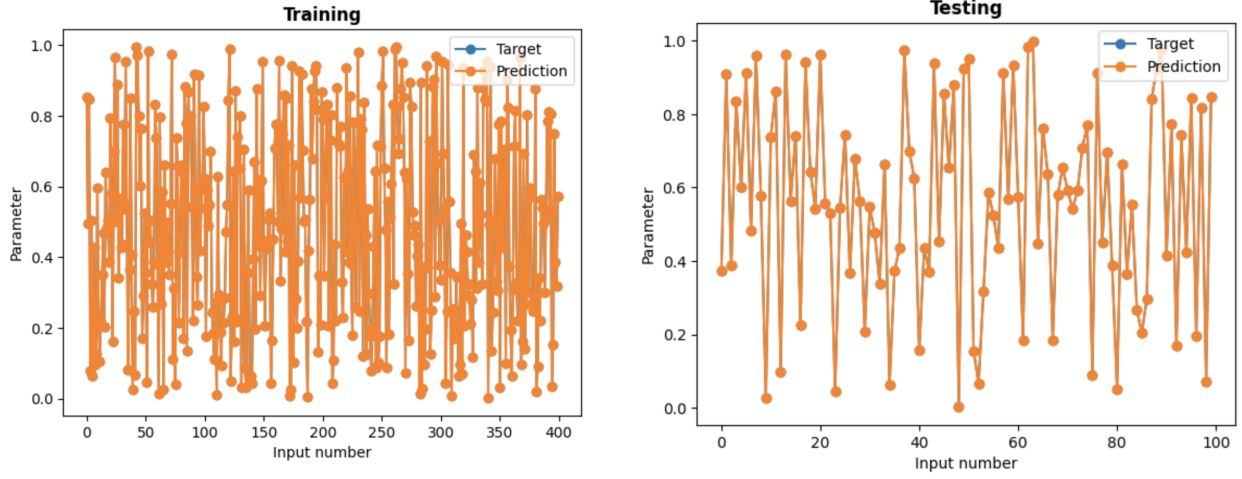


Figure 2: Training and testing stages for Werner states.

The prediction is really good, with a MSE^1 of 10^{-19} . To obtain this result, we have used a reservoir of $N = 6$ spins.

3.2 Cross states

After this first example, we move on to more complex 2-qubit states. Instead of one parameter, we now define the so called cross-states, which depend on 3 independent parameters. Their general density matrix is:

$$\rho_C = \begin{pmatrix} p & 0 & 0 & c_1 \\ 0 & q & c_2 & 0 \\ 0 & c_2 & q & 0 \\ c_1 & 0 & 0 & p \end{pmatrix}, \quad \text{with } p + q = 1, c_1 \leq |p|, c_2 \leq |q|. \quad (3.3)$$

The last two conditions on the non-diagonal elements come from the general requirement ($\rho_{14} \leq \sqrt{\rho_{11}\rho_{44}}$ and $\rho_{23} \leq \sqrt{\rho_{22}\rho_{33}}$) for the density matrix to represent a physical state.

In this case, it will be necessary to insert the input multiple times to obtain accurate results. That is, instead of measuring the reservoir after the first evolution, we take the partial trace again over all spins but the first two, and reinsert the input on the first two qubits. Then we let the reservoir evolve again, and repeat this process as many times as necessary.

3.2.1 Entanglement metrics

In order to characterize the level of entanglement of a bipartite system like ours, we use two different metrics: concurrence and negativity, both defined for a two-qubit state ρ [5].

- **Concurrence.** Defined as:

$$C(\rho) := \max(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4), \quad (3.4)$$

where $\lambda_1, \dots, \lambda_4$ are the eigenvalues, in decreasing order, of the following Hermitian matrix:

$$R = \sqrt{\sqrt{\rho} \tilde{\rho} \sqrt{\rho}}, \quad (3.5)$$

and

$$\tilde{\rho} = (\sigma_y \otimes \sigma_y) \rho^* (\sigma_y \otimes \sigma_y). \quad (3.6)$$

¹Computed over a set of 100 input states in the testing phase by comparing the difference between their actual λ parameter and the prediction.

- **Negativity.** Defined as the absolute sum of the negative eigenvalues of the partially transposed density matrix:

$$\mathcal{N}(\rho) = \sum_i \frac{|\lambda_i| - \lambda_i}{2} \quad (3.7)$$

If the negativity is 0, the state is separable. On the other hand, for entangled states it is < 0 .

The classification task was seen to work much better when using negativity as a metric, so in this report we will not comment the results obtained with the concurrence. We want to classify the states in entangled or separable, without predicting directly any entanglement metric or parameter. For this purpose, we train the reservoir by assigning the vector $Y = [1, 0]$ to separable states, and $Y = [0, 1]$ to entangled states. In the testing phase, the reservoir will provide as output a 2D vector. If the first element of this vector is larger than the other, the input state is classified as separable. Otherwise, it is considered to be entangled.

3.2.2 Results

The next figures show the evolution of the classification accuracy as the number of input reinsertions is increased. We see that inserting the input multiple times in fact improves the performance of the algorithm, which makes no mistakes with 3 iterations. This has been seen to be the tendency for this family of two-qubit states. Figure 6 shows the performance of the algorithm when asked to predict the negativity of the input states directly, with 3 input insertions. The MSE in this case is of the order of 10^{-3} .

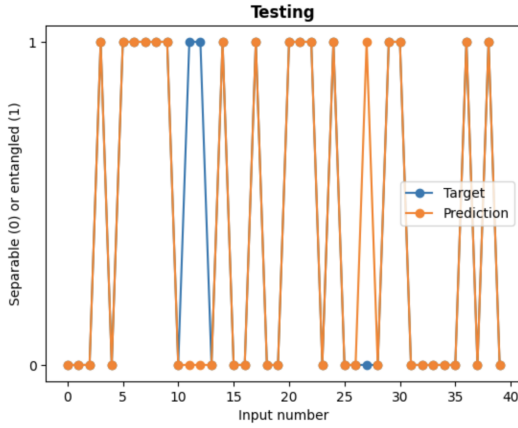


Figure 3: 1 iteration: 3 mistakes.

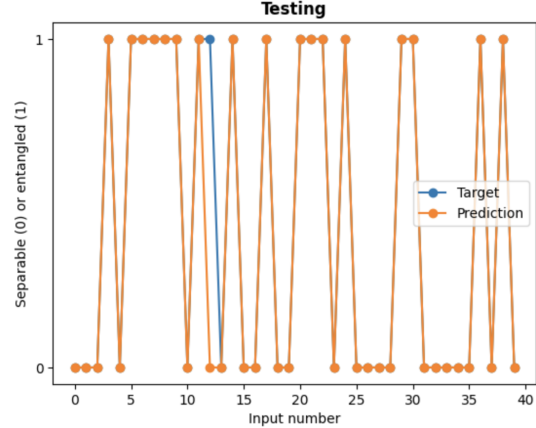


Figure 4: 2 iterations: 1 mistake.

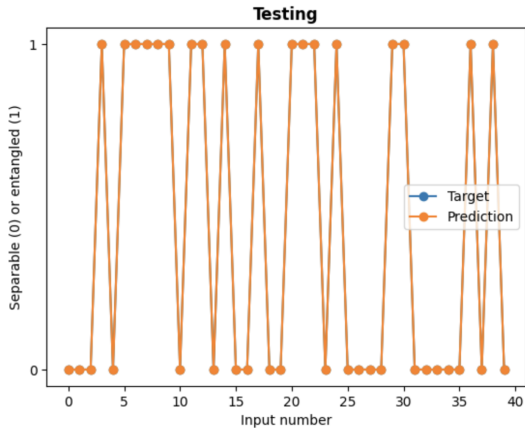


Figure 5: 3 iterations: no mistakes.

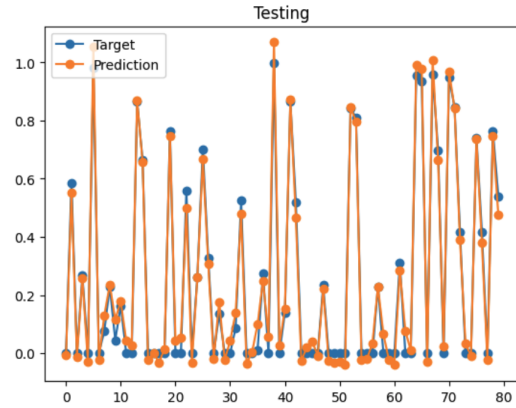


Figure 6: Prediction of negativity for 3 iterations.

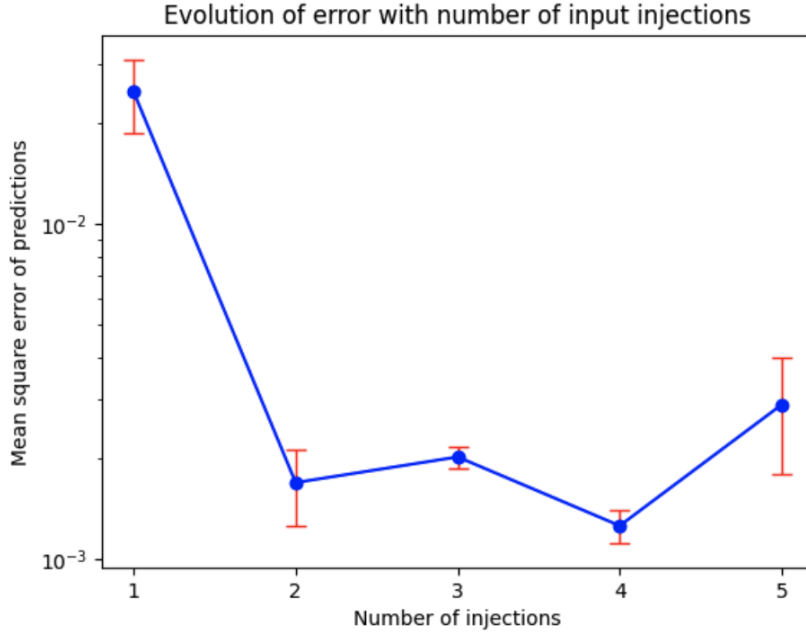


Figure 7: Evolution of error with number of inputs.

In order to generalize the previous observations, we perform a statistical average of the MSE for the negativity prediction over multiple random seeds (which modifies the coupling matrix of the Hamiltonian, and thus the structure of the reservoir lattice). Figure 7 shows the evolution of this average with the number of input reinjections. As seen previously, reinserting the input improves the performance, 4 times being the optimal number. Beyond this number, the MSE seems to increase.

4 Conclusions

This work has shown that the memory of the reservoir turns out to be useful for classification tasks which are not temporal in nature. This is true at least for the entanglement classification task under study. Further work could consist in analyzing the performance of this scheme with other families of quantum states, in order to generalize the conclusions achieved thus far. It has also become clear that the metric used to quantify entanglement plays an important role in the accuracy of the classification. It would also be interesting to study which properties make a metric stand out, and if this advantage depends on the underlying quantum system of the reservoir. Additionally, not much attention was paid to the hyperparameters associated to the Ising lattice, like the intensity of the transverse magnetic field or the range of values of the elements in the coupling matrix. A more thorough inspection could have found values for which the algorithm worked better.

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